

3-Minute Presentation Script: Emotion Detection

0:00 - 0:30 | Introduction & Problem

(Visuals)

Good morning everyone. I am Om Vyas, and today I am presenting my project: a Real-Time Emotion Detection System.

Emotion recognition is vital for modern tech like adaptive learning systems and mental health monitoring. However, most existing solutions require heavy processing power, complex installations, or paid cloud APIs. My goal was to build a completely free, highly optimized solution that runs entirely in the browser using any standard webcam, with zero setup required by the user.

0:30 - 1:15 | Tech Stack & Architecture

(Visuals)

To achieve this, I built the web application using Streamlit and integrated streamlit-webrtc to capture live video directly within the browser without any local drivers.

The application works on a frame-by-frame pipeline. First, OpenCV's Haarcascade rapidly detects and crops the user's face. Then, that cropped image is passed to a custom PyTorch Convolutional Neural Network, which processes and classifies the emotion into one of seven categories like Happy, Sad, or Angry. This entire cycle is highly optimized and happens in under 100 milliseconds per frame.

1:15 - 1:45 | The CNN Model

(Visuals)

The brain of the system is the CNN, which I trained from scratch on the FER-2013 dataset, containing over 35,000 face images.

By utilizing data augmentation, Batch Normalization, and Dropout layers, the model generalizes well and avoids overfitting. It successfully handles the high variance of human expressions and varying lighting conditions found in this dataset.

1:45 - 2:30 | Live Demonstration

(Visuals)

As you can see in the working application, the system is incredibly responsive. The bounding box and emotion labels update seamlessly as my expressions change.

Additionally, the UI features a live analytics dashboard on the side. This tracks the frequency of detected emotions over the last 60 seconds, creating a continuous emotional profile that provides more value than just frame-by-frame guesses.

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2:30 - 3:00 | Conclusion & Future Scope

(Visuals)

To conclude, this project proves that we can run complex AI vision models smoothly and efficiently in a standard web environment.

Looking ahead, my future enhancements include adding voice-based emotion detection for a multi-modal approach, and converting the model to run natively on mobile devices.

Thank you for your time. I am now open to any questions.